Network layer (Basics)

Now lets come to the 3rd layer — Network layer.

If you remember we did talk about transport layer providing process to process communication, now network layer provides host to host communication.

- Network layer exists in:
  - * Routers
  - END devices.

Services & protocols:

→ transport segments from sending to receiving hosts.

Sender: encapsulates segments into datagrams & passes to link layers.

Receiver: Delivers ~~data~~ segments to transport layer protocol.

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Protocol: IP

network layer has one protocol only - IP.

→ In all previous layers we also only had end devices but now we also have Routers. as Intermediate devices.

Routers: • examine header fields in all IP datagrams passing through it.

• Then it moves them from input to correct output ports to transfer datagrams along end-end path.

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2 key Functions:

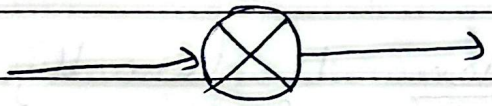
We already know these both:

1) Routing
2) forwarding.

1) Forwarding:

Local per router function

Transferring a packet from a router's incoming link interface to the appropriate outgoing link interface based on the packet's header.



Forwarding:

* Handled in hardware using forwarding table

"DATA PLANE"

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2) Routing:

Network-wide Action

Determining the end-end path that packets take from the source host to the destination host.

~~Routing~~ executed in Software using Routing Algorithms.

"Control Plane"

Data Plane & Control Plane:

Dataplane: Is basically that decides forwarding.

Control Plane: network-wide logic to Routing.

There are 2 control plane approaches:

1) traditional Routing algos:
implement the software in every router.

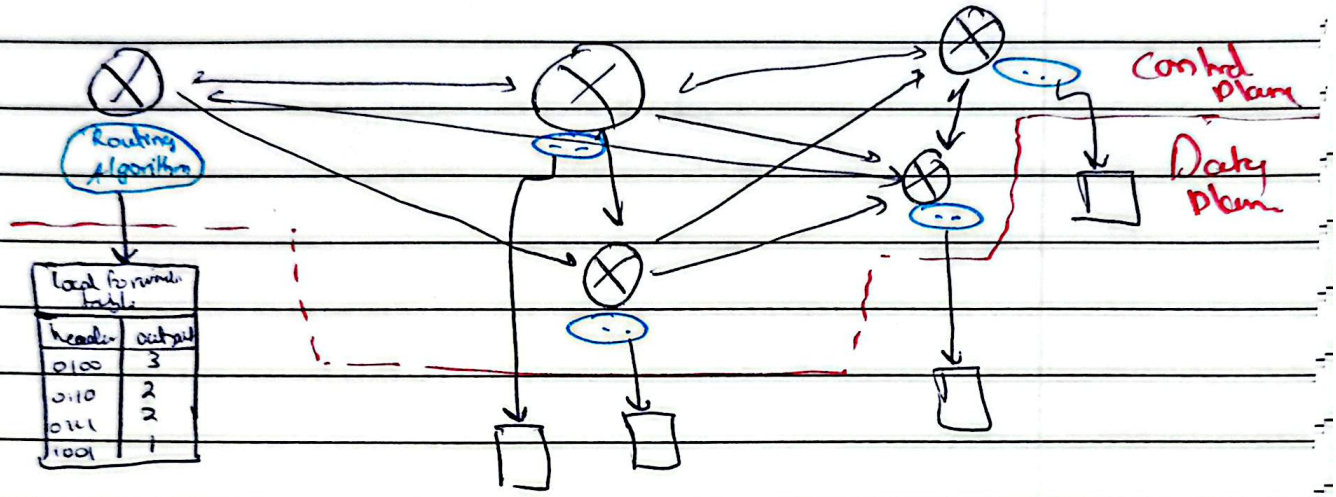
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2) Software Defined Networking (SDN):

implemented in remote servers.

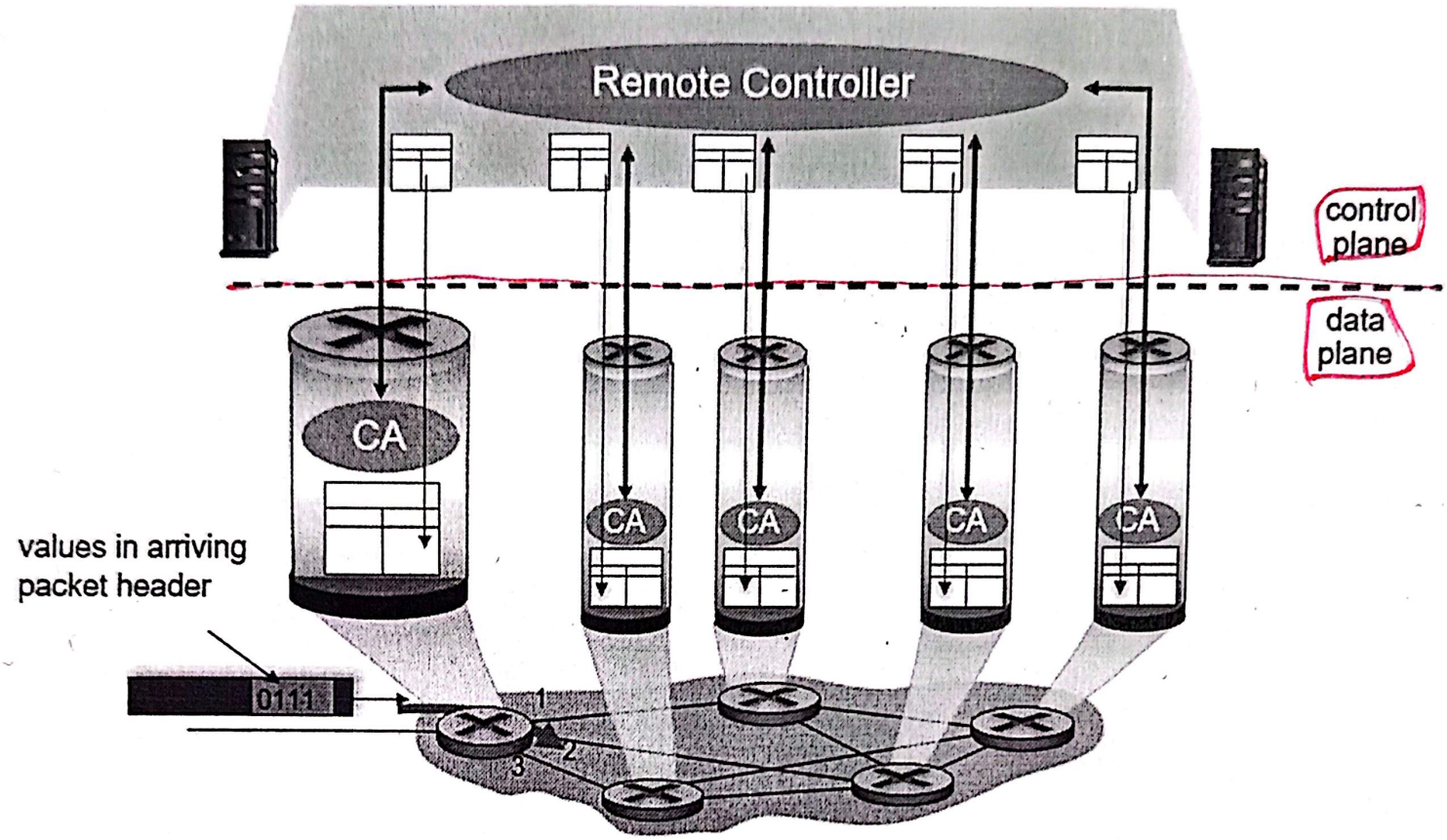
1) Traditional approach:



In this every router has a routing Algorithm & each & every router interact with the control plane.

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2) SDN approach:



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Network Service Model:

What service model should be used for transporting datagrams & what do they offer?

- e.g - consider a service for a single datagram:
  - we can guarantee delivery & very low delay:
- consider services for a flow of datagram:
  - we can guarantee in-order delivery & minimum bandwidth to flow.

Is it possible to get best of everything? e.g bandwidth, loss, order, ~~cost~~ timing.

For this we have many models:

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1) Best effort model:

No guarantee on:
- → sucessful datagram delivery to destination
- → timing or order of delivery.
- → bandwidth available to end-end flow.

But the best-effort model has huge success:
- → simplicity of mechanism allows internet to widely deployed.
- → sufficient bandwidth allows good performance of realtime apps easily.
- → congestion control.

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Some extra detail:

Before Internet IP model we had ATM model (Asynchronous Transfer Model).
ATM used to be connection-oriented unlike IP being connectionless.
AND ATM didn't use packet switching & instead divided data into fixed 53 byte sized cells.

ATM guaranteed Bandwidth, no loss of cell/packet, & best Quality of Service.
But even though it was so good it could not scale.

2) Constant bit rate.
→ Its architecture is ATM.

3) Available bit rate

4) Intserv Guaranteed

5) Diffserv

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Network Architecture	Service Model	Quality of Service (QoS) Guarantees ?			
		Bandwidth	Loss	Order	Timing
Internet	best effort	none	no	no	no
ATM	Constant Bit Rate	Constant rate	yes	yes	yes
ATM	Available Bit Rate	Guaranteed min	no	yes	no
Internet	Intserv Guaranteed (RFC 1633)	yes	yes	yes	yes
Internet	Diffserv (RFC 2475)	possible	possibly	possibly	no

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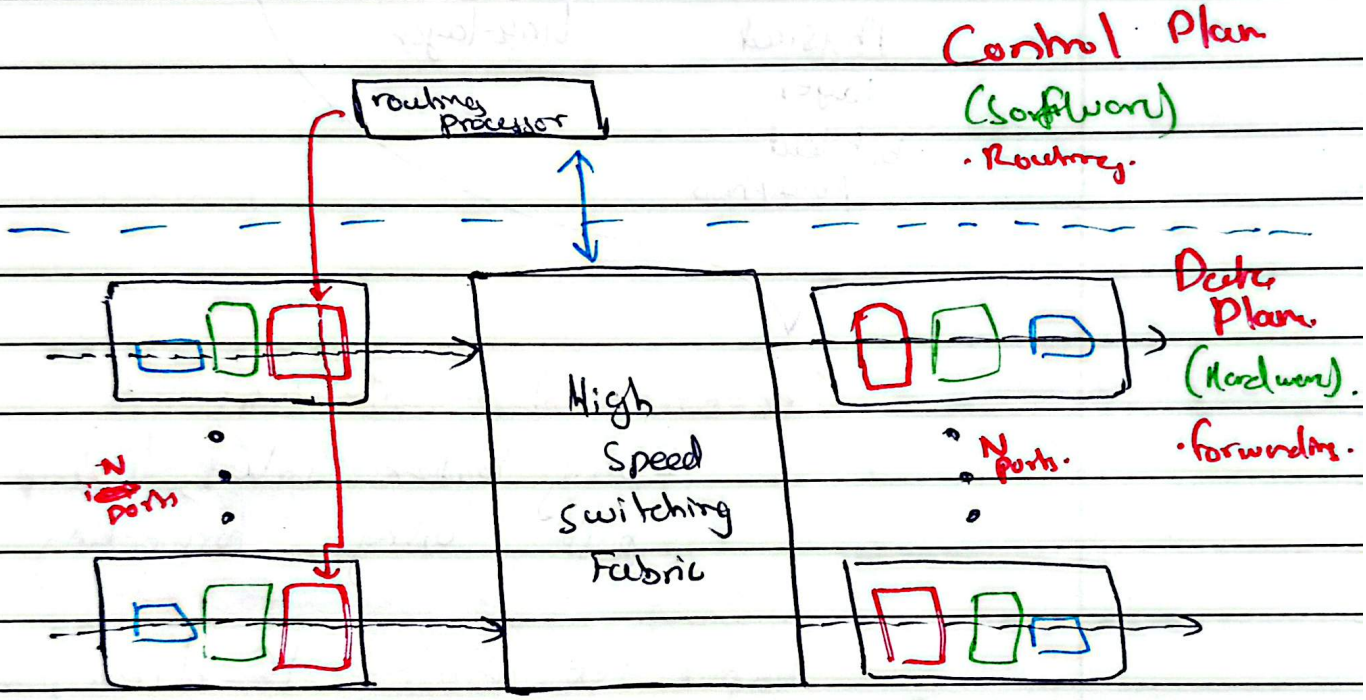
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Network layer
Data Plane: Inside a Router

Now we will understand Dataplane & Routers especially & see what is actually inside Routers that allows forwarding.

Router Architecture.



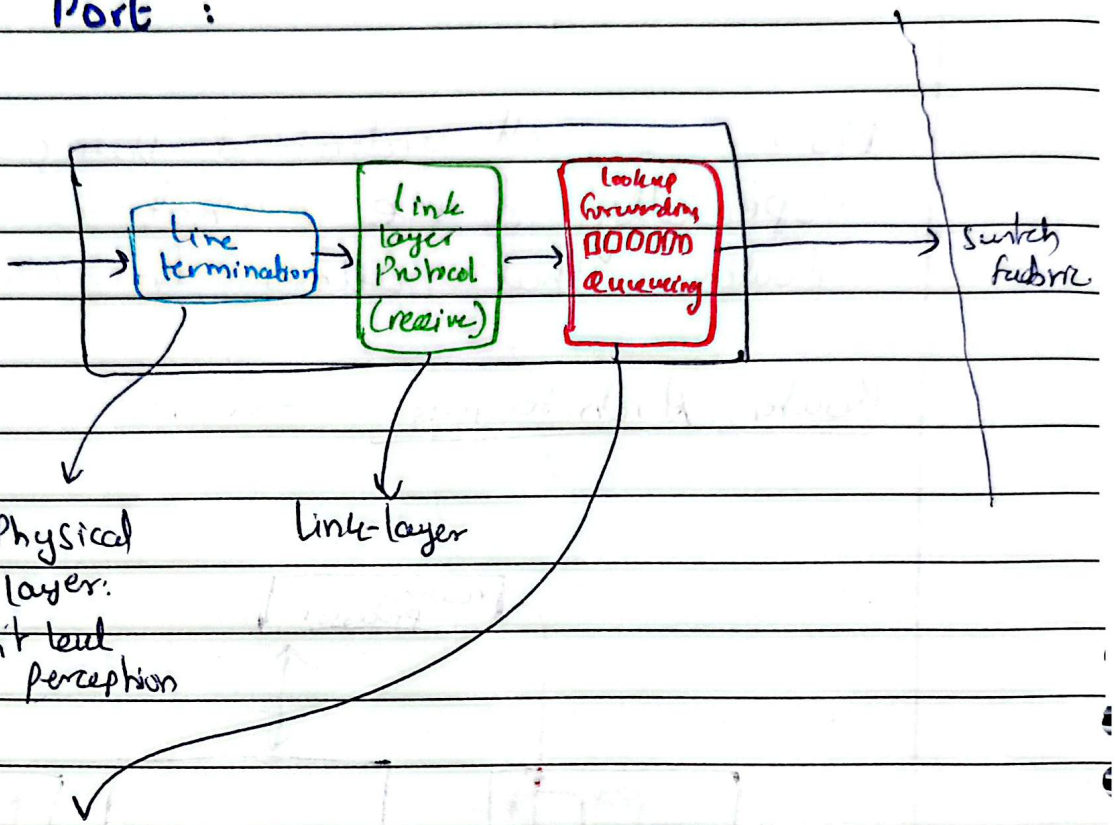
- Like we have said before too, the Data Plane ~~ha~~ has forwarding which is done by hardware that operates in nanoseconds time frame

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Input Port :



Decentralized switching:

- Using header values, lookup output port using forwarding table

Input Port Queuing. If datagrams arrive faster than forwarding rate into switch fabric.

Destination - based - forwarding:

forward using destination IP address

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generalized forwarding: Forward based on any set of header field values.

we will only learn Destination based forwarding.

Destination - Based forwarding

In this we see the destination Address bits & choose ranges for a particular output.

e.g

Destination Address Range	Link interface
11001000 00010111 00010000 00000000 to 11001000 00010111 00011111 11111111	0
11001000 00010111 00011000 00000000 to 11001000 00010111 00011000 11111111	1
Other wise	2

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But a small problem comes:

what if ranges don't divide up this easily.

e.g

64000 - 112000	0
70000 - 70100	1

} - overlapping ranges.

To prevent overlap we can divide the first range into 2 but that would mean a very big forwarding table.

So to solve ~~this~~ it we keep this configuration but use longest prefix matching.

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Longest Prefix matching:
When looking for forwarding table entry given destination address, use longest adress prefix that matches destination address.

e.g.

Destination address Range	Link interface
11001000 00010111 00010*** ********	0
11001000 00010111 00011000 ********	1
11001000 00010111 00011*** ********	2
Otherwise	3

now

e.g

11001000 00010111 00010110 10100001

matches interface 0. ✓ chosen.

11001000 00010111 00011000 10101010

interface 2.

Interface 1, so it is chosen.

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11001000 00010111 00011001 00000006

Interface 1 matches till here but after this it does NOT match so — interface 1 cannot be taken.
It should be an exact match except for "*".

So interface 2.

Longest Prefix matching is used & performed using TCAM, (Ternary content addressable memory).
these retrieve address in one clock cycle regardless of table size.

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Input port: Important:

bits ⟶ frames ⟶ datagrams ⟶ switching fabric.
(physical layer) (link layer) (network layer)

It converts the bits to datagrams / Packets so we can see network headers & send to particular output link (where its again converted to bits).

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*Switching Fabric.

Okay so now we have the packet from the input ports & now we need to redirect it to the particular output port using the info given by input port (it tells where to go using the forwarding table + longest prefix matching).

Switching fabric just need to move the datagrams packets from input to output.

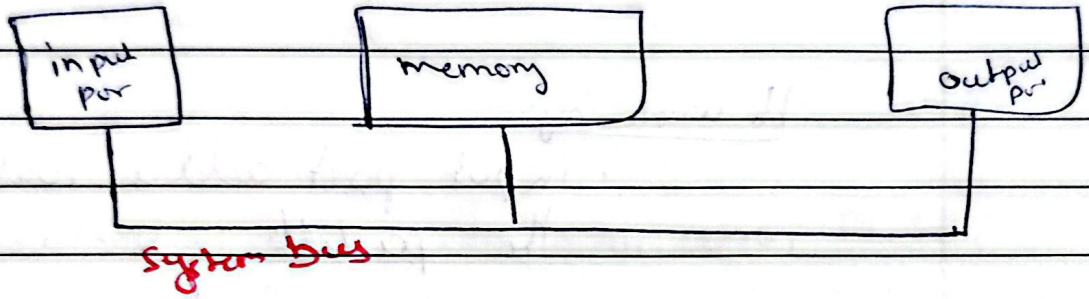
Switching rate: rate at which packets can be transferred. from input to output

- Remeber that there are N input ports & all connected to a switching fabric so best rate ideally is N times the line rate. , NR.

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3 major types of switching fabrics:

1) Switching via memory:



In start, routers were like computers, an incoming packet from input port would trigger an interrupt to CPU. The packet will then be moved to main memory using bus. Then the routing processor looks up output port & copies packet from main memory to output port.

Cons:

- Only one bus so no parallel work
- every packet travels on bus twice so Bus bandwidth/2.

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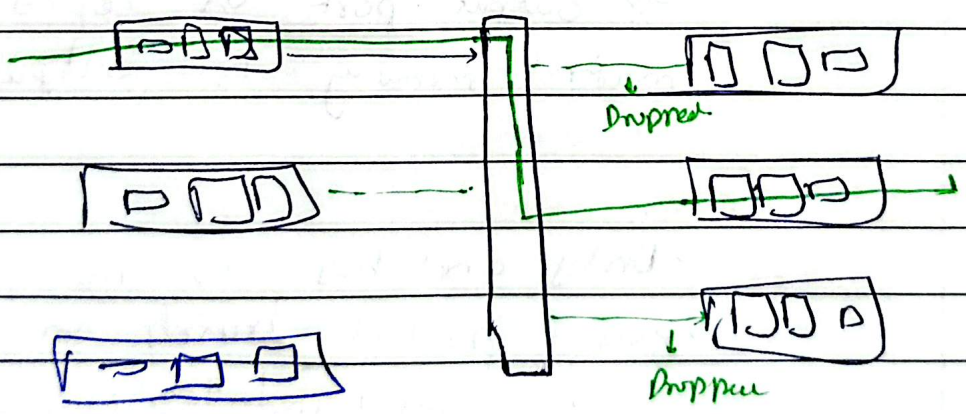
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2) Switching via Bus:

Input ports transfer packets directly to output ports over a shared bus without intervention of CPU.

It work by:

- input port adds a routing tag to the packets indicating output port & transmits into the bus.
- All outputs receive it but only the port matching tag keeps it other drop it.
- The routing tag is then stripped off



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Con:

Because only one bus link is shared only one packet data time can traverse.

Router switching speed is capped by bus speed.

3) Switching via Interconnection Network.

To overcome single bus limitation, modern routers use interconnected network such as a cross bar switch.

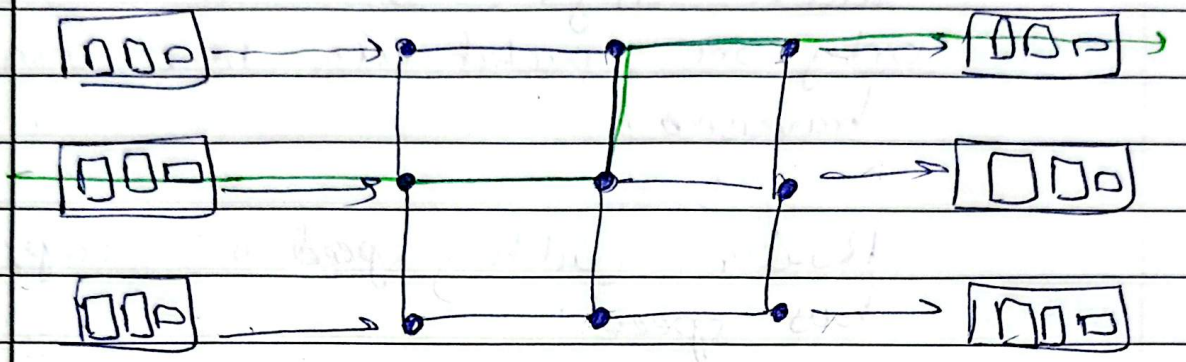
It works:

- A cross bar consisting of N inputs buses intersecting with N output buses in a 2D grid matrix.
- At each intersection, a switching controller can open or close an electronic switch.

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Typically the used to be 3×3 matrix but then they also became too less so it became an 8×8 matrix.

The size of matrix depends upon input Ports N & N output Ports.

But as N increased, the matrix is N^2 so it separated, and it wasn't possible to handle such a big cross bar.

So instead Multiple switching plans started to be used.

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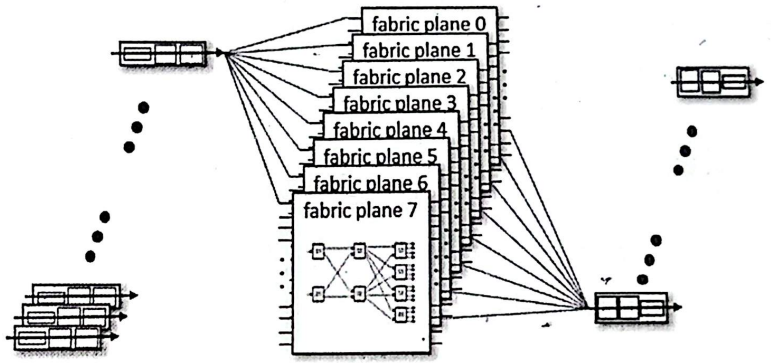
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Like multiple crossbars on top of each other.

- scaling, using multiple switching "planes" in parallel:
  - speedup, scaleup via parallelism

- Cisco CRS router:
  - basic unit: 8 switching planes
  - each plane: 3-stage interconnection network
  - up to 100's Tbps switching capacity



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Input port Queuing:

This is a problem that happens when

$$N * R_{line} < R_{switch}$$

R_{line} = input link transmission speed.

R_{switch} = switch rate limit

so if our switching rate is lower than line transmission rate then Queuing starts to happen.

Another reason can be: multiple packets across different input ports need to go to same output port, a bottleneck forms.

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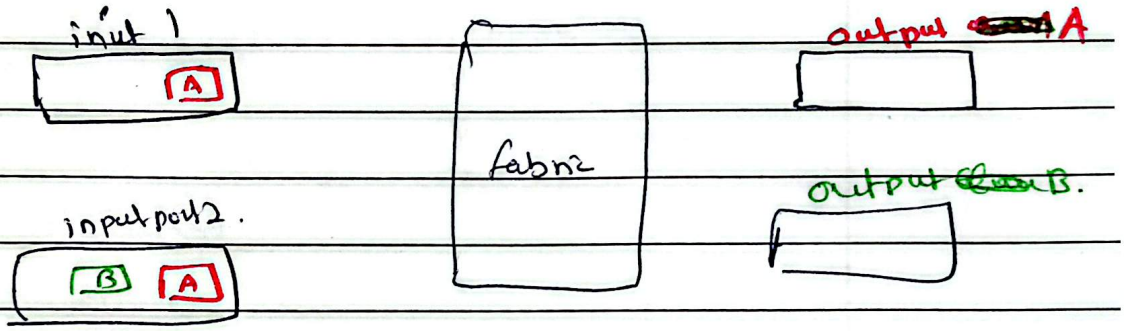
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Head-of-line Blocking:

A queued packet at the front of an input queue prevents packets behind it from being forwarded even if the destination output ports for those trailing packets are completely free & idle.

consider scenario



- now both input ports get a packet A at same time. but consider input port 1's is chosen first
Now even though our Output B is free & the Packet B in input Port 2 is blocked bcz of packet A in front of it.

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- Because of HOL input queues build up rapidly even at moderate traffic loads.

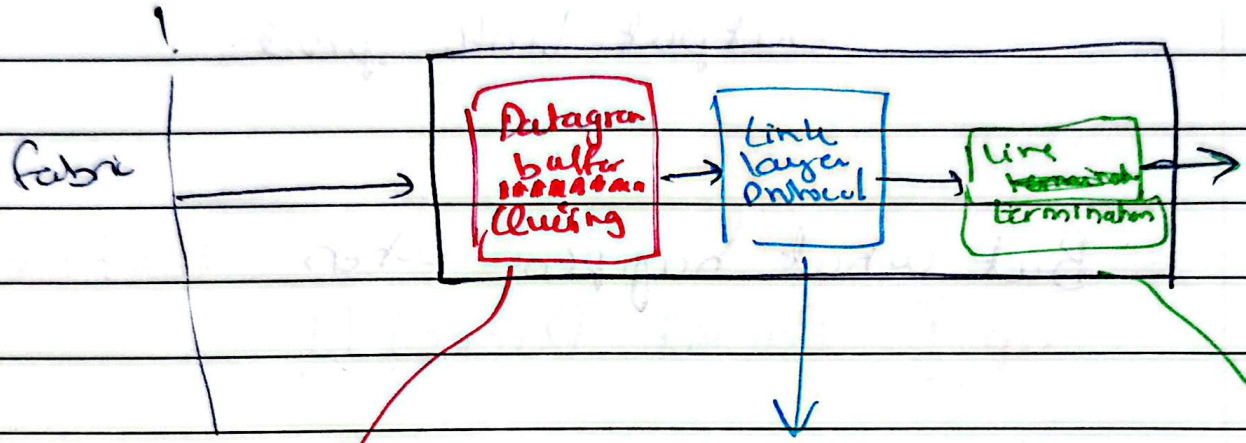
- If queue becomes full, Packet can be loss.

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Output port:

Output port is just like input port but opposite



Stores packets delivered by switching fabric if they arrive faster than transmission rate of output link

Encapsulates IP datagram to link-layer frames.

Convert frames bits into physical signals.

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Output Port Queuing:

Buffering: This happens when datagrams arrive faster from fabric than output link speed.

But what happens when **Buffer fills up?** Packet Drops!!!

But how to choose which Packet to Drop.

It follows a **Drop policy.**

Buffer management

- e.g. If buffer is full drop the new packet. (Tail drop)
- Instead of waiting for buffer to fill completely, track queue length & begin dropping packets (priority wise). (Priority Drop). or just mark them to signal congestion to end devices.

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How to decide the size of Buffer of output ports?

Traditional way:

$$B = RTT \cdot C$$

B = buffer size

RTT = Typical Round trip time

C = Link Capacity (bandwidth) of router's output interface

Modern way:

$$B = \frac{RTT \cdot C}{\sqrt{N}}$$

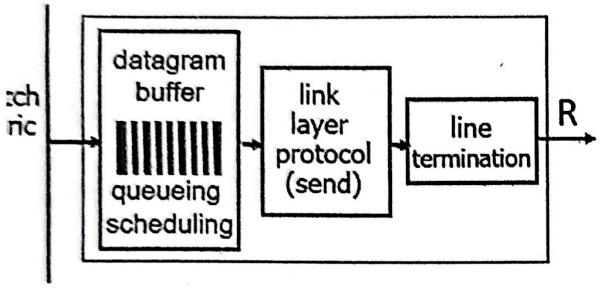
N = number of unique independent flows / no. packets currently flowing through the router.

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• Too much buffering can increase delays.

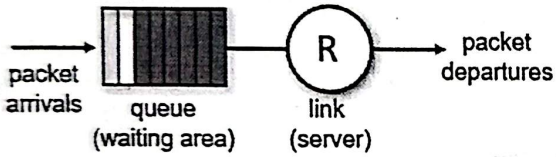
Buffer Management



buffer management:

- drop: which packet to add, drop when buffers are full
  - tail drop: drop arriving packet
  - priority: drop/remove on priority basis
- marking: which packets to mark to signal congestion (ECN, RED)

Abstraction: queue



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Packet Scheduling Policies:

Now for the output buffer, it has to decide which packet to send next.

For this we have multiple policies:

- FCFS (First come, First served).
- Priority
- Round Robin
- weighted fair queuing.

Packet Scheduling: FCFS

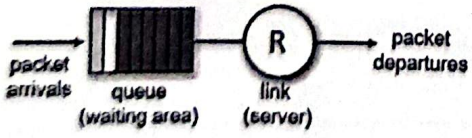
packet scheduling: deciding which packet to send next on link

- first come, first served
- priority
- round robin
- weighted fair queueing

FCFS: packets transmitted in order of arrival to output port

- also known as: First-in-first-out (FIFO)
- real world examples?

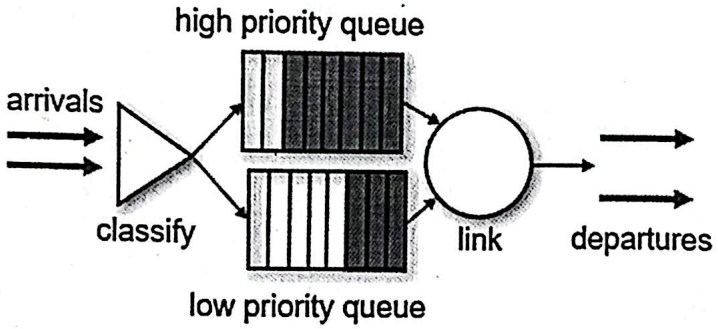
Abstraction: queue

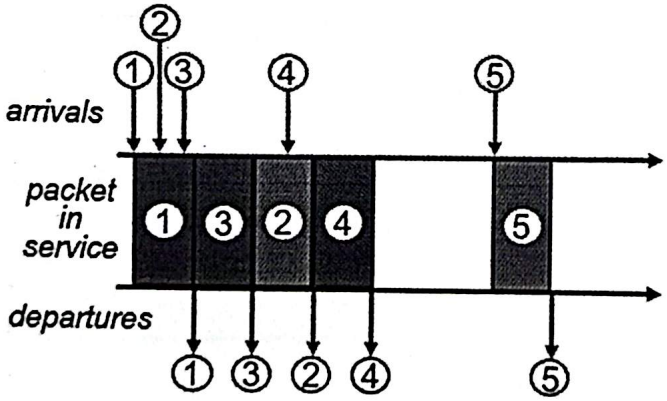


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2) Priority:

- Arriving traffic is classified & queued by class.
- Any header field can be used for classification.
- FCFS within a priority class.



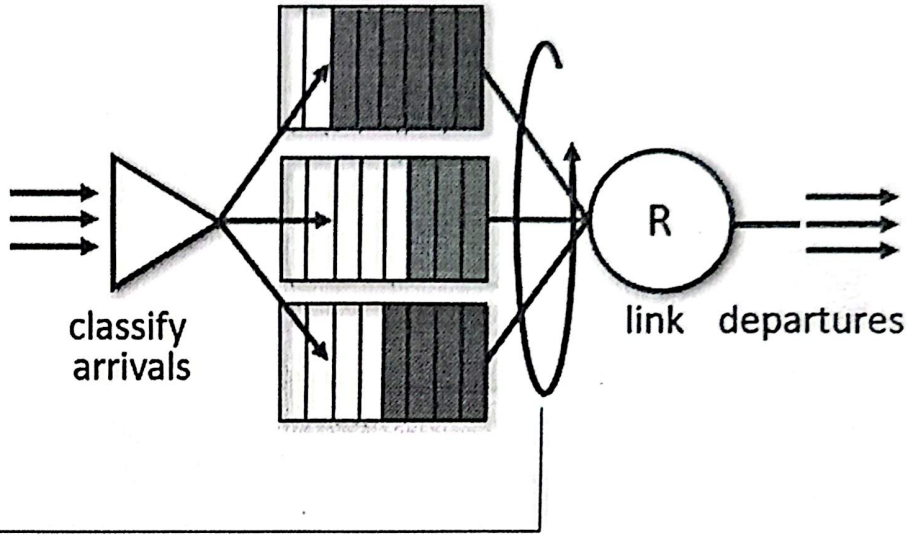


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3) Round Robin

Round Robin (RR) scheduling:

- arriving traffic classified, queued by class
  - any header fields can be used for classification
- server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn



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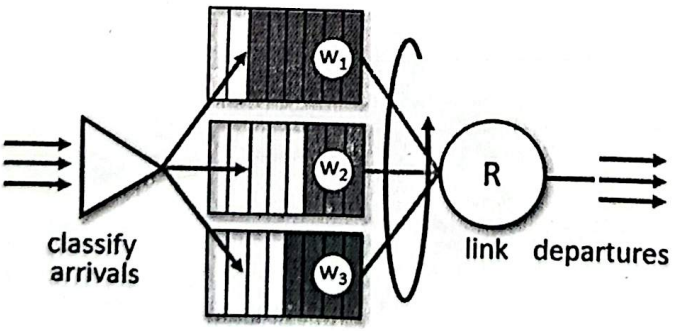
4) Weighted Fair queueing:

This is advanced version of Round Robin.

- Each class i is assigned a weight.
- During any time where all queues have packets class i is guaranteed to receive a fraction of total link bandwidth equal to:

$$\frac{w_i}{\sum w_i}$$

If link speed is R, class i is guaranteed throughput: $R * \frac{w_i}{\sum w_i}$



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